# Midterm Test
# F18CE – Multivariable Calculus
# Test 2 2025/26

**4 March 2026**

**Duration: 30 minutes**

**No calculators allowed**

1. A twice differentiable function $f : \mathbb{R}^2 \to \mathbb{R}$ is defined by

$$f(xy, y^2) = \tfrac{1}{3}(y - x).$$

Let $F(x, y) = f(u, v)$, where $u : \mathbb{R}^2 \to \mathbb{R}$ and $v : \mathbb{R} \to \mathbb{R}$ are defined by $u = xy$ and $v = y^2$.

(a) Suppose $y \neq 0$. Find the expression for $f_{uv}(u, v)$ and $f_{vu}(u, v)$. Hence, determine whether $f_{uv}(u, v) = f_{vu}(u, v)$. **[5]**

(b) Does the equation $F(x, y) = \sin(xy)$ determine the existence of a function $y = y(x)$ near $(0, 0)$? Provide a full justification. **[4]**

2. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined as $f(x, y) = e^{-x-y}$.

(a) Find $P_2(x, y)$, the second-degree Taylor polynomial of $f$ around the point $(0, 0)$. Hence, deduce $P_n(x, y)$, the $n$th degree Taylor polynomial of $f$ around $(0, 0)$. **[4]**

(b) Write down the error term $R_2(x, y)$ and hence deduce $R_n(x, y)$. **[3]**

ANSWER SCHEME

## Question 1.

(a) $f(u,v) = f(xy, y^2) = \frac{1}{3}(y - x)$ (1). $u = xy, v = y^2 \Rightarrow u_x = y, u_y = x, \frac{dv}{dy} = 2y$.

$f, f_u, f_v$ / \ $u$ $v$ / \ | $x$ $y$ $y$

$\frac{\partial}{\partial x}(1) : f_u u_x = -\frac{1}{3} \Rightarrow 3yf_u = -1$ (2) (½ mark)

$\frac{\partial}{\partial y}(2) : 3f_u + 3y(f_{uu}u_y + f_{uv}\frac{dv}{dy}) = 0 \Rightarrow f_u + xyf_{uu} + 2y^2 f_{uv} = 0$ (1 mark)

$\Rightarrow f_{uv} = -\frac{1}{2y^2}(f_u + xyf_{uu}), y \neq 0$ (½ mark)

$\frac{\partial}{\partial y}(1) : f_u u_y + f_v \frac{dv}{dy} = \frac{1}{3} \Rightarrow 3xf_u + 6yf_v = 1$ (3) (½ mark)

$\frac{\partial}{\partial x}(3) : 3f_u + 3xf_{uu}u_x + 6yf_{vu}u_x = 0 \Rightarrow f_u + xyf_{uu} + 2y^2 f_{vu} = 0$ (1 mark)

$\Rightarrow f_{vu} = -\frac{1}{2y^2}(f_u + xyf_{uu}), y \neq 0$ (½ mark)

$\therefore f_{uv}(u,v) = f_{vu}(u,v)$ (1 mark)

(b) Let $G(x,y) = 3\sin(xy) - y + x \Rightarrow G(0,0) = 0 \Rightarrow (0,0)$ is a solution point. (1 mark)

$G_y(x,y) = 3x\cos(xy) - 1 \Rightarrow G_y(0,0) = -1 \neq 0$. (2 marks)

Since $G_x = 3y\cos(xy) + 1$ and $G_y$ exist and continuous for all points, it follows that $G$ is a function of class $C^1$, and there exists a solution point $(0,0) \in \mathbb{R}^2$ such that $G_y(0,0) \neq 0$. Then by the implicit function theorem, there exists an explicit function $y = y(x)$ near $(0,0)$ such that $y(0) = 0$. (1 mark)

## Question 2

(a) $f(x,y) = e^{-x-y} \Rightarrow \begin{cases} f_x(x,y) = f_y(x,y) = -f(x,y)\,* \\ f_{xx}(x,y) = f_{yy}(x,y) = f_{xy}(x,y) = f(x,y)\,** \end{cases}$ (1 mark)

$\Rightarrow f(0,0) = 1 \Rightarrow \begin{cases} f_x(0,0) = f_y(0,0) = -1\,* \\ f_{xx}(0,0) = f_{yy}(0,0) = f_{xy}(0,0) = 1\,** \end{cases}$ (1 mark)

for all order *odd/**even partial derivatives.

$P_2(x,y) = 1 - x - y + \frac{1}{2}(x^2 + y^2 + 2xy) = 1 - (x+y) + \frac{1}{2}(x+y)^2$ (1 mark)

$P_n(x,y) = 1 + \sum_{r=1}^{n} \frac{(-1)^r}{r!}(x+y)^r$ (1 mark)

(b) $R_2(x,y) = \frac{1}{2}\left(x^2 f_{xx}(\tilde{x},\tilde{y}) + y^2 f_{yy}(\tilde{x},\tilde{y}) + 2xyf_{xy}(\tilde{x},\tilde{y})\right)$

$= \frac{1}{2}\left(x^2 e^{-\tilde{x}-\tilde{y}} + y^2 e^{-\tilde{x}-\tilde{y}} + 2xye^{-\tilde{x}-\tilde{y}}\right)$

$= \frac{1}{2}e^{-\tilde{x}-\tilde{y}}(x+y)^2$ (1 mark)

$R_n(x,y) = \frac{(-1)^n}{n!}e^{-\tilde{x}-\tilde{y}}(x+y)^n$, (1 mark)

where $\tilde{x}$ is in between $x$ and 0; $\tilde{y}$ is in between $y$ and 0. (1 mark)